

MANICRAFTS TECHNOLOGY

TASK 1: Fundamentals of Automobile Systems and Introduction to Electric Vehicles

1. What is an Automobile?

An automobile is a self-propelled vehicle used for transporting people or goods from one place to another. It converts energy into mechanical motion to move the vehicle. Automobiles can be powered by:

- Internal Combustion Engines (ICE)
- Electric Motors (EVs)
- Hybrid Systems

Automobiles play an important role in transportation, trade, and daily life.

2. Major Systems of a Conventional Automobile

a) Engine System

The engine is called the heart of an automobile. It converts fuel energy into mechanical power.

Functions:

- Produces power for vehicle movement.
- Converts chemical energy of fuel into mechanical energy.

Types:

- Petrol Engine
- Diesel Engine

b) Transmission System

The transmission system transfers power from the engine to the wheels.

Components:

- Clutch
- Gearbox
- Differential
- Drive Shaft

Functions:

- Controls speed and torque.
- Provides smooth power transmission.

c) Braking System

The braking system slows down or stops the vehicle.

Types:

- Drum Brake
- Disc Brake
- Anti-lock Braking System (ABS)

Functions:

- Ensures safety.
- Controls vehicle speed.

d) Steering System

The steering system controls the direction of the vehicle.

Types:

- Manual Steering
- Power Steering

Functions:

- Changes vehicle direction.
- Improves maneuverability.

e) Suspension System

The suspension system absorbs shocks from road irregularities.

Functions:

- Provides ride comfort.
- Maintains vehicle stability.
- Reduces vibration.

f) Electrical System

The electrical system supplies electrical power to different components.

Components:

- Battery
- Alternator
- Lights
- Sensors

Functions:

- Starting the engine.
- Operating electrical accessories.

3. Introduction to Electric Vehicles (EVs)

An Electric Vehicle (EV) is a vehicle powered by an electric motor using energy stored in rechargeable batteries.

Features:

- Uses electric motors for propulsion.
- Stores energy in batteries.
- Produces zero tailpipe emissions.

- Reduces dependence on fossil fuels.

Examples:

- Electric Cars
- Electric Scooters
- Electric Buses

4. Major Components of an Electric Vehicle

a) Battery Pack

The battery pack stores electrical energy required to power the vehicle.

Features:

- Usually Lithium-Ion based.
- Main energy source of EV.

b) Electric Motor

The electric motor converts electrical energy into mechanical energy.

Types:

- BLDC Motor
- PMSM Motor
- Induction Motor

c) Inverter

The inverter converts DC power from the battery into AC power for the motor.

Functions:

- Controls motor speed.
- Improves efficiency.

d) Battery Management System (BMS)

The BMS monitors battery performance.

Functions:

- Monitors voltage and temperature.
- Prevents overcharging.
- Ensures battery safety.

e) Charger

The charger replenishes battery energy.

Types:

- On-board Charger
- Off-board Charger

f) Controller

The controller regulates power flow between battery and motor.

Functions:

- Controls acceleration.
- Manages vehicle performance.

5. Working Principle of EV

Step 1: Battery supplies electrical energy.

Step 2: Inverter converts DC power into AC power.

Step 3: Motor receives electrical power and rotates.

Step 4: Mechanical power is transmitted to wheels.

Step 5: Vehicle moves forward.

6. Comparison Between ICE and Electric Vehicles

Parameter	ICE Vehicle	Electric Vehicle
Energy Source	Petrol/Diesel	Electricity
Emissions	High	Zero Tailpipe Emissions
Noise	High	Very Low
Maintenance	High	Low
Efficiency	Low	High
Running Cost	High	Low

7. Advantages of Electric Vehicles

1. Environment Friendly.
2. Lower Running Cost.
3. Reduced Maintenance.
4. Quiet Operation.
5. Higher Energy Efficiency.
6. Reduced Dependence on Fossil Fuels.

8. Challenges of Electric Vehicles

1. Limited Charging Infrastructure.
2. High Battery Cost.
3. Long Charging Time.
4. Range Anxiety.
5. Battery Recycling Challenges.

9. Current Trends in EV Industry

Government Policies

Governments worldwide are promoting EV adoption through subsidies, tax benefits, and charging infrastructure development. India supports EVs through schemes such as FAME.

EV Manufacturers and Startups

Major EV manufacturers include:

- Tesla
- Tata Motors
- Mahindra Electric
- BYD
- Ola Electric
- Ather Energy

Growth of Charging Infrastructure

Public charging stations are increasing rapidly, making EV usage more convenient.

Future Scope of EV Technology

- Fast Charging Technology
- Solid-State Batteries
- Wireless Charging
- Autonomous Electric Vehicles
- Improved Battery Life and Efficiency

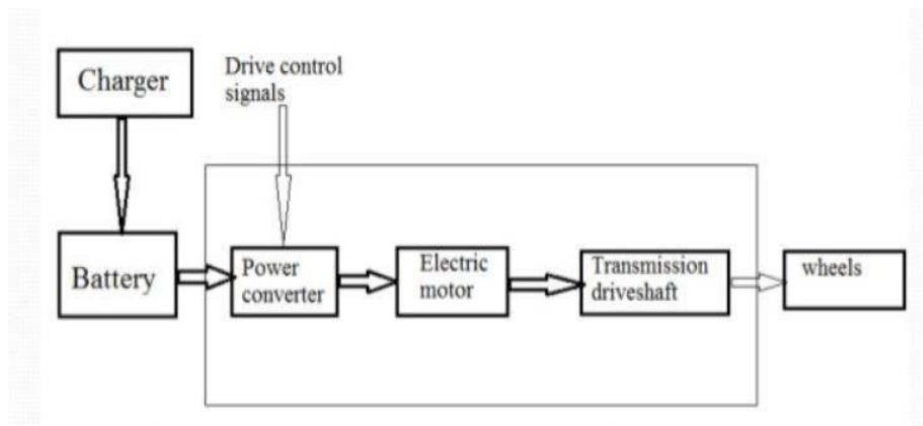


Fig.1.Electric Vehicle System

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